Fiscal Year:	FY 2024	**Task Last Updated:**	FY 11/29/2024
PI Name:	Desai, Rajeev Ph.D.		
Project Title:	Research Review: Neurobehavioral Biomarkers Performance/Pathway		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02478	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	02/19/2019	**End Date:**	09/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	End date changed to 09/30/2025 per L. Juliette/JSC (Ed., 11/7/24). End date changed to 06/30/2024 per L. Juliette/JSC (Ed., 9/27/23).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kangas, Brian Ph.D. (Harvard Medical School) Limoli, Charles Ph.D. (University of California, Irvine)		
Grant/Contract No.:	Directed Research		
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Task Description:	Deep space missions to Mars and beyond involve significant challenges for astronauts including extended exposure to spaceflight stressors such as galactic cosmic rays (GCR), altered gravity, social isolation/confinement, and prolonged sleep disruption. These spaceflight stressors, alone or in combination, can alter neurobiological function at multiple levels in the brain to impair neurocognitive performance and jeopardize mission success. Thus, we first documented the existing evidence of how the four major spaceflight stressors -- including radiation, microgravity, isolation/confinement, and sleep deprivation -- alone or in combination, alter molecular, neurochemical, and neurobiological signatures that may be linked to operationally relevant neurobehavioral and neurocognitive performance in a research review. Over the course of this data mining effort, we found that limited information is currently available on how exposure to spaceflight stressors alone or when combined alters multiple neurotransmitters in key brain regions involved in complex neurocognitive function. Given that neurotransmitters in the brain provide a direct readout of the central nervous system (CNS) function at multiple levels -- including behavioral, emotional, stress, and cognitive function -- it is critical to fully map how exposure to space-relevant spaceflight stressors, alone and when combined, alters temporal and regional neurotransmitter activity in vivo, and whether the observed neurochemical alteration in the brain impacts neurocognitive performance. Thus, in subsequent studies, we first examine how exposure to two major spaceflight stressors alone, (space radiation or sleep fragmentation) alters neurotransmitter networks in key brain regions that play an important role in complex neurocognitive processes, and whether these changes can be associated with other neurobiological and peripheral biomarkers. These supplemental studies were conducted in freely moving mice to examine the impact of acute or chronic exposure to simulated 33-beam GCRSim or 6-hr or 10-hr 5 days/week of sleep fragmentation on multiple neurotransmitters that may govern complex neurocognitive function. We anticipate that results from these studies will provide a solid foundation for future work designed to delineate the effects of other spaceflight stressors alone (e.g., social isolation/confinement or microgravity) and when combined, on neurotransmitter function and neurocognitive consequences. Aim 1: Integrated research review for identifying neurobehavioral biomarkers impacted by spaceflight stressors -- including radiation, altered gravity, sleep deprivation, and isolation/ confinement. Overall goal is to document the existing evidence of how the four major spaceflight stressors -- including radiation, microgravity, isolation/confinement, and sleep deprivation-- alone or in combination, alter molecular, neurochemical, and neurobiological signatures that may be linked to operationally relevant neurobehavioral and neurocognitive performance. Consider a broad array of existing research from laboratory animals to humans on how acute and long-term exposure to spaceflight stressors, alone or in combination, impacts behavior and cognitive function and whether such effects can be attributed to multimodal changes in the brain. Where appropriate, repurpose knowledge from other CNS-health studies to astronauts (e.g., aging, disorder, disease). Summarize current literature on the impact of spaceflight stressors, alone or in combination, on critical peripheral biomarkers that potentially could be associated with behavioral and cognitive deficits. Document open questions and identified research gaps in the knowledge base that connects molecular, neurochemical, and neurobiological pathways to operational performance and provide recommendations for future multimodal research on spaceflight stressors. Aim 2: Neurotransmitters in the brain provide a direct readout of CNS function at multiple levels -- including behavioral, emotional, stress, and cognitive function. It is critical to fully map how exposure to space-relevant radiation alters temporal and regional neurotransmitter activity in vivo and whether such changes in the brain impact neurocognitive performance. Here, we conducted studies in mice to examine the impact of the complex 33-beam terrestrial GCR simulation on prefrontal cortex (PFC) and ventral tegmental area (VTA) neurotransmitters involved in translational touchscreen-based assays of reward sensitivity/motivation and sustained attention. Mapping acute and chronic GCRSim-induced alterations in neurotransmitter networks in a key brain neurocognitive performance pathway (i.e., projection from VTA to PFC) that is involved in complex neurocognitive function will accelerate NASA's ability to develop evidence-based strategies for countering the detrimental consequences of space radiation on the brain and neurocognitive performance. Aim 3: Using a multimodal approach, assess impact of 6-hr or 10-hr 5 days/week sleep disturbance on a) neurochemical signatures in the PFC and VTA; b) complex behavior performance on economic demand, Psychomotor Vigilance Test (PVT), fear conditioning, novel object location memory (OLM)/updating task and monitoring behavioral activity (this battery of tasks varies in difficulty and addresses different aspects of behavior and complex neurocognitive function); c) neurobiological signatures using high resolution anatomy, diffusion tensor imaging (DTI), and functional connectivity; d) molecular endpoints of synaptic function, including inflammatory markers; and e) peripheral plasma biomarkers, including a full metabolomic and lipidomic analysis using Liquid Chromatography-Mass Spectrometry (LC-MS/MS). The overarching goal of these studies is to determine whether alterations in neurotransmitter biomarkers that cause neurocognitive deficits is associated with changes in molecular and brain activity that can be linked to changes in peripheral metabolite and lipid biomarkers.
Rationale for HRP Directed Research:	This project is directed because it contains highly constrained research. This study will 1) expand the biomarker database through identifying neurotransmitters and neurobehavioral biomarkers to elucidate effects of both acute and chronic simulated GCR or sleep deprivation relative to controls, and 2) identify biomarkers that will best indicate Combined Behavioral Stressor (CBS) performance decrements in ground animal model of space radiation and sleep deprivation.
Research Impact/Earth Benefits:	
	Aim 1: Here, we completed our datamining research reviews and submitted two final reports to HFBP. In addition, information generated from these two research review was published in several articles in peer-reviewed journals between 2021 to 2023 (see below). Task Completed/Progress: 1) Final research review/datamining report and presentation entitled "Nonhuman Primate Central Nervous System Radiation/Performance Data Mining" submitted to NASA-HFBP 2019. 2) Final research review/datamining report and presentation entitled "Neurobehavioral Biomarkers Performance/Pathway" submitted to NASA-HFBP in 2020. Article in Peer-reviewed Journals: 1) Desai RI, Kangas BD, Limoli CL. Nonhuman primate models in the study of spaceflight stressors: Past contributions and future directions. Life Sci Space Res (Amst). 2021 Aug;30:9-23. PMID: 34281669. doi: 10.1016/j.lssr.2021.03.008. 2) Alwood JS, Mulavara AP, Iyer J, Mhatre S, Rosi S, Shelhamer M, Davis-Takács C, Dinges D, Mao XW, Desai RI, Elgart SR, Whitmire A, Williams TJ. Summary Report for the Technical Interchange Meeting: Circuits and Biomarkers of the Central Nervous System Relating to Astronaut Performance. 2021. NASA STI Program, Mail Stop 148, NASA Langley Research Center, Hampton, VA, 23681-2199. National Technical Information Service, 5285 Port Royal Road, Springfield, VA, 22161.

	This report is also available in electronic form at http://www.sti.nasa.gov/ and http://ntrs.nasa.gov 3) Desai RI, Limoli CL, Stark CEL, Stark SM. Impact of spaceflight stressors on behavior and cognition: A molecular, neurochemical, and neurobiological perspective. Neurosci Biobehav Rev. 2022 Jul;138:104676. doi: 10.1016/j.neubiorev.2022.104676. Epub 2022 Apr 22. PMID: 35461987. 4) Alwood JS, Mulavara AP, Iyer J, Mhatre SD, Rosi S, Shelhamer M, Davis C, Jones CW, Mao XW, Desai RI, Whitmire AM, Williams TJ. Circuits and Biomarkers of the Central Nervous System Relating to Astronaut Performance: Summary Report for a NASA-Sponsored Technical Interchange Meeting. Life (Basel). 2023 Aug 31;13(9):1852. PMID: 37763256; PMCID: PMC10532466. doi: 10.3390/life13091852. Aim 2: In these studies, we determined neurotransmitter changes using in vivo microdialysis procedures in freely moving mice with LC-MS/MS analysis from the PFC and VTA in mice exposed to complex 33-beam terrestrial acute or chronic GCR Sim vs controls. Neurotransmitter network data will be related to neurocognitive (economic demand and Psychomotor Vigilance Test, PVT) data in these same mice. We report the first findings of how the PFC neurotransmitter networks are reorganized by realistic complex GCR simulations and how that informs neurocognitive outcomes to provide a foundational and comprehensive outlook regarding the impact of space radiation on CNS function.
Task Progress:	Task Completed/Progress: Data for neurocognitive tasks have been collected, analyzed, and completed. All neurotransmitter data for this task has been collected and analysis of samples from PFC has been completed. These data are now published (see below). In addition, neurotransmitter data from the VTA have been analyzed and these data are now being prepared for manuscript submission (see below). Article in Peer-reviewed Journals: 1) Desai RI, Kangas BD, Luc OT, Solakidou E, Smith EC, Dawes MH, Ma X, Makriyannis A, Chatterjee S, Dayeh MA, Muñoz-Jaramillo A, Desai MI, Limoli CL. Complex 33-beam simulated galactic cosmic radiation exposure impacts cognitive function and prefrontal cortex neurotransmitter networks in male mice. Nat Commun. 2023 Nov 27;14(1):7779. PMID: 38012180; PMCID: PMC10682413. doi: 10.1038/s41467-023-42173-x Article in Preparation: 1) Yu IL, Smith EC, Dawes MH, Burke EL, Solakidou E, Ma X, Kangas BD, Makriyannis A, Chatterjee S, Dayeh MA, Muñoz-Jaramillo A, Desai M, Limoli CL, Desai RI. Complex 33-beam simulated galactic cosmic radiation exposure impacts ventral tegmental area neurotransmitter networks in male mice. Aim 3: In this work, we used a multimodal approach to determine how exposure to 6-hr or 10-hr sleep fragmentation alters molecular, neural, and neurotransmitter activity in the brain of male vs female mice to impact behavioral and neurocognitive performance as well as to identify any peripheral biomarkers that can be related to changes in CNS function. Task Completed/Progress: Data for all endpoints in male mice have been collected, and analysis has been completed for behavioral and neurocognitive task, neurotransmitter levels, neural signatures, and molecular endpoints. Analysis is ongoing for metabolites/lipid biomarkers. Thus far, data from male mice show that sleep fragmentation produces differential effects on behavior and neurocognitive performance. For example, 6-hr or 10-hr of sleep fragmentation impacts motivation, attention, and reward sensitivity, but does not produce changes in assays that provide a measure of fear/anxiety and memory. At the molecular level, while some changes in inflammatory markers were observed, overall sleep fragmentation does not significantly alter synaptic or inflammatory markers in the brain. At the neurotransmitter level, data suggests that sleep fragmentation selectively impacts the monoamines dopamine and norepinephrine in the PFC, but not other neurotransmitters. At the neural activity level, data thus far suggests, that sleep fragmentation restricts gray matter water diffusivity in key brain regions (e.g., cortex, basal ganglia, midbrain, and cerebellum), which may contribute to a reduction in perivascular clearance prolonging retention of metabolic waste and unwanted proteins. Analysis of peripheral biomarkers in male mice is ongoing. Given that plasma metabolite or lipid biomarkers represent a useful approach amenable for real-time in-flight biomarker assessment. These easily obtainable bio-samples can be quickly and longitudinally analyzed to assess developing health problems and to understand any impact on behavior and cognitive performance. In addition, behavioral and neurocognitive task, neurotransmitter levels, neural signatures, and molecular endpoints data in female mice are currently being collected. These studies will be completed in the upcoming project period. In summary, this multimodal approach will directly address NASA's Human Research Program (HRP) Integrated Research Plan (IRP) program. Data obtained from these studies will provide a clear link between neurobiological changes in the brain at multiple levels and its impact on translationally relevant neurocognitive performance. Identifying these changes will facilitate NASA's ability to develop effective strategies to counter the negative consequences of spaceflight stressors.
Bibliography Type:	Description: (Last Updated: 01/23/2026)
Articles in Peer-reviewed Journals	Alwood JS, Mulavara AP, Iyer J, Mhatre SD, Rosi S, Shelhamer M, Davis C, Jones CW, Mao XW, Desai RI, Whitmire AM, Williams TJ. "Circuits and biomarkers of the central nervous system relating to astronaut performance: Summary report for a NASA-sponsored technical interchange meeting." Life (Basel). 2023 Aug 31;13(9):1852. https://doi.org/10.3390/life13091852 ; PMID: 37763256; PMCID: PMC10532466 , Aug-2023
Articles in Peer-reviewed Journals	Desai RI, Kangas BD, Luc OT, Solakidou E, Smith EC, Dawes MH, Ma X, Makriyannis A, Chatterjee S, Dayeh MA, Muñoz-Jaramillo A, Desai MI, Limoli CL. "Complex 33-beam simulated galactic cosmic radiation exposure impacts cognitive function and prefrontal cortex neurotransmitter networks in male mice." Nat Commun. 2023 Nov 27;14(1):7779. http://dx.doi.org/10.1038/s41467-023-42173-x ; PMID: 38012180; PMCID: PMC10682413 , Nov-2023

NASA Technical Documents	Alwood JS, Mulavara AP, Iyer J, Mhatre S, Rosi S, Shelhamer M, Davis-Takács C, Dinges D, Mao XW, Desai RI, Elgart SR, Whitmire A, Williams TJ. "Nervous System Relating to Astronaut Performance." Houston, Texas: NASA Johnson Space Center, 2021. NASA/TP-20210016273. https://ntrs.nasa.gov/api/citations/20210016273/downloads/TP-20210016273%20-%20combined.pdf, Jun-2021